

The Sovereign Agentic AI Stack: Control, Choice, and Open Architectures

Vincent Caldeira
CTO APAC, Red Hat



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Sovereignty & Trust Deficits in Proprietary AI



Extreme Silicon Concentration

A single chip designer holds 80-95% market share, presenting a severe hardware chokepoint and single point of failure across global supply chains.



Geopolitical Fragmentation

Over 90% of global AI data center capacity resides under US and China control, exposing enterprise operations to aggressive trade controls and decoupling.



The Inference Bottleneck

Legacy L4/L7 networking load-balancing fails because standard round-robin cannot scale for stateful Key-Value (KV) cache memory constraints.



The Token Economy

Cloud operational spend scales uncontrollably. Workloads remain highly sensitive to latency and unpredictable billing cycles tied directly to proprietary APIs.



The Open Answer: Sovereignty Through Open Architectures



Autonomy is Superior to Isolation

True sovereignty is not built on isolationism or siloed engineering. It is the capacity to act deliberately within an interdependent ecosystem while avoiding vendor lock-in.

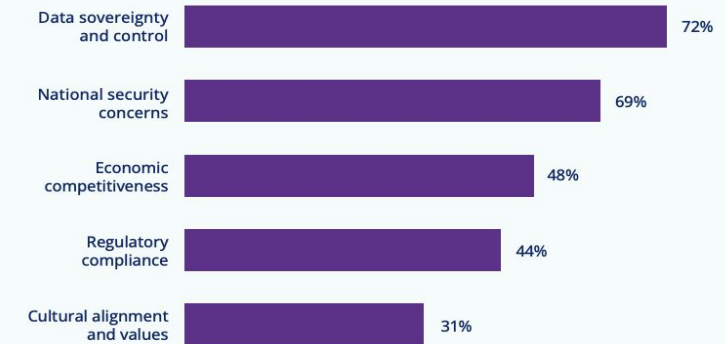


The Open Source Architecture

Coordinating standard-driven architectures to decouple software from underlying silicon, transforming sovereignty from a barrier into a foundation for innovation.

KEY INTEREST DRIVERS IN SOVEREIGN AI

In your opinion, what is driving interest in Sovereign AI? (select all that apply)



2025 Global Collaboration in AI Survey, Q16,
Sample Size = 233, Total Mentions = 607, DKNS excluded (3%)

79%

of organizations view
Sovereign AI as a strategic
priority

82%

developing customized AI
to maintain control over
capabilities & IP

93%

view global collaboration
as essential for secure
sovereign AI



The 3-Layer Sovereign Stack: A Bottom-Up Overview

Unifying Standalone Tools into a Connected, Multi-Cloud Operational Fabric



Layer 3: Agentic Interaction & Governance (AAIF Standards) [MCP](#) [A2A Protocol](#) [AGENTS.md](#) [Goose](#)

- Standardizes agent-to-tool (MCP) and agent-to-agent (A2A) interactions under neutral Linux Foundation governance [AAIF].
- Decouples agent cognitive reasoning from proprietary host environments and vendor APIs.



Layer 2: Cloud-Native Substrate & Governance (CNCf Orchestration) [Kubernetes](#) [KServe](#) [Istio Fabric](#) [Envoy AI Gateway](#) [vCluster / DRA](#)

- Offloads operational tasks—token rate-limiting, KV-cache routing, and confidential sandboxing—from Python app code into platform primitives.
- Enforces zero-trust workload identity (SPIFFE) and multi-tenant GPU isolation.



Layer 1: Distributed Compute & Hardware Abstraction (PyTorch Foundation) [PyTorch Core](#) [Ray](#) [vLLM Engine](#) [Triton](#) [OpenReg](#)

- Unifies model training, distributed orchestration, and high-throughput inference serving under open governance.
- Delivers cross-architecture enablement (NVIDIA CUDA, AMD ROCm, Intel, NPUs) to eliminate silicon lock-in.

The Strategic Architecture Shift

Moving away from fragmented, ad-hoc Python wrappers and proprietary API rentals toward a connected, vendor-neutral operational fabric that guarantees sovereignty across any cloud and accelerator.



Layer 1 (Compute): PyTorch Foundation & Hardware Abstraction

Eliminating Silicon Lock-In via an Open, Cross-Architecture Execution Substrate

Multi-Project Open Governance

PyTorch CorevLLMRayDeepSpeedSafetensors

- Unifies deep learning training, distributed orchestration, and high-throughput inference serving under neutral Linux Foundation governance.
- Serves as the de facto industry execution standard, ensuring day-0 compatibility for open-weight foundation models.

Hardware Abstraction & Kernel Compilers

OpenRegTritonHelion DSLvLLM-cpu

- **OpenReg Interface:** Out-of-tree device registration allowing new silicon accelerators to plug directly into PyTorch without custom code forks.
- **Portable Compilers:** Triton and Helion DSL deliver state-of-the-art kernel performance across heterogeneous hardware without rewriting model code.
- **CPU Execution:** vLLM-cpu enables inference on standard CPUs (Intel Xeon AMX) to operate within power-constrained data centers.

Heterogeneous Silicon Choice

NVIDIA CUDAAMD ROCmIntel XPUGoogle TPURegional NPU

- Makes hardware selection a runtime deployment decision rather than an architectural constraint.
- Delivers day-0 enablement for diverse AI accelerators and domestic NPUs to preserve hybrid-cloud compute agility.

PyTorch Foundation Framework & Serving Stack

PyTorch • vLLM • Ray • DeepSpeed

Hardware Abstraction & Compiler Layer

OpenReg Interface • Triton • Helion DSL

Heterogeneous Accelerator Pool

NVIDIA CUDA • AMD ROCm • Intel XPU • Google TPU • Domestic NPUs



Layer 2 (Platform): Cloud-Native AI Substrate

TIER 1

Agentic & Developer Portal

Declarative templating and automated CI/CD integration workflows for AI engineers.

Backstage

kagent

TIER 2

Traffic & API Governance

Model virtualization, token-aware rate limiting, and centralized JWT/IRSA credential exchange.

Envoy AI Gateway

Gateway API

TIER 3

MLOps & Delivery

End-to-end model training pipelines, cryptographic OCI signing, and immutable packaging.

Kubeflow

KServe

Harbor

Kitops

TIER 4

Dynamic Orchestration

Gang scheduling, topology-aware accelerator placement, distributed inference co-scheduling, and multi-tenant hierarchical queuing

Volcano

Kueue

llm-d

TIER 5

Execution & Hardware

Device-level GPU partitioning (MIG), kernel isolation, and hardware-enforced sandboxes.

NVIDIA GPU Operator

DRA

gVisor / Kata



Cloud Native Agentic Standards



Cloud Native Agentic Workload Definition — Microservice-oriented, containerized workloads on Kubernetes that leverage AI reasoning to plan, execute actions, and govern their own execution based on event-driven signals.

Pillar I

Identity & Tenancy

Workload Identity: Cryptographically verifiable unique identities per agent.

Continuous Auth: Verify identity dynamically before every autonomous action.

Data Isolation: JIT access & query-level segregation on shared vector stores.

Pillar II

Security & Tools

Anti-Hijacking: Restrict execution to pre-approved schemas & boundaries.

Prompt Defense: Prevent system prompt leakage & sanitize internal flows.

Hardened Containers: Non-root execution & signed OCI provenance.

Pillar III

Observability

MELT Standard: Correlate telemetry metrics & traces for explainability.

AI Telemetry: Track GPU usage, token economics, & LLMOps lifecycle.

Agent-as-Judge: Automated evaluation and tamper-resistant audit pipelines.

Pillar IV

Fault Tolerance

API Extensions: Path-based routing and custom inference endpoints.

K8s Resilience: Topology constraints & PodDisruptionBudgets.

Elastic Autoscaling: Dynamic HPA driven by inference & GPU pressure.

Lifecycle Scope: Vendor-neutral architectural governance across Kubernetes cluster environments

<> Develop

🔗 Distribute

🏠 Deploy

▶ Runtime

Layer 3: Standardizing Agent Interoperability via AAIF Open Standards

Establishing open, vendor-neutral protocol standards to prevent fragmentation and enable agent-to-tool and agent-to-agent communication.



GOVERNANCE & ECOSYSTEM

The AAIF Neutral Governance Layer

Neutral Ecosystem Home

AAIF (under Linux Foundation) stewards foundational open protocols:

[MCP](#) [A2A](#) [AGENTS.md](#) [Goose](#)

Vendor-Neutral Interoperability

Decouples agent reasoning from host infrastructure to prevent lock-in.

Prevents vendor lock-in across execution runtimes.



CORE OPEN PROTOCOLS

Protocol Architecture: MCP + A2A

A2A (AGENT-TO-AGENT)

HORIZONTAL LAYER



Discovers capabilities, negotiates tasks, & coordinates cross-domain workflows.

MCP (AGENT-TO-TOOL)

VERTICAL LAYER ('USB-C')



Universal standard connecting agents vertically to enterprise tools & data sources.

Protocol Synergy: A2A routes agent handoffs; MCP executes precise tool calls within agents.



VALUE & GOVERNANCE

Strategic Interoperability Outcome



Portable & Modular

Workloads interoperate across any framework, cloud, or enterprise boundary.

Cross-Framework Multi-Cloud



Standardized Governance

Uniform schema validation, identity mapping, and auditable tracing across all agent handoffs.

- Uniform schema validation
- Identity mapping & auth propagation
- End-to-end auditable tracing

Data & Model Sovereignty: Why Sovereign Infrastructure Needs Open Models

Achieving true autonomy requires moving beyond sovereign hardware to open weights, auditable supply chains, and local domain adaptation

The Risk of Infrastructure Without Model Autonomy

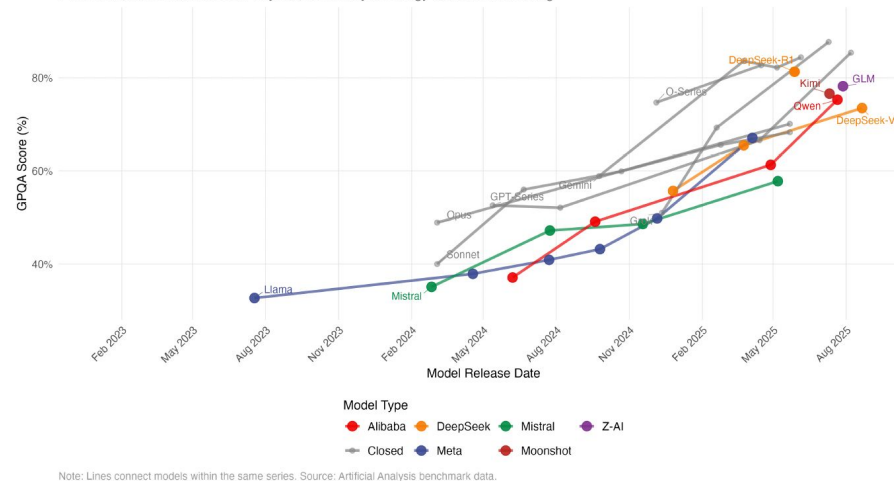
- **Proprietary Monopoly:** Closed model APIs capture 96% of revenue and 80% of usage, exposing workloads to vendor lock-in.
- **Black-Box Opacity:** Users cannot inspect proprietary pipelines, creating risks around copyright, data provenance, and backdoors.
- **Cultural & Contextual Gap:** Mass-market models underperform on localized languages, regional regulations, and domain contexts.

The Convergence Advantage: Open Models Catch Up in 3–6 Months

Open-weight models match proprietary benchmark performance with a rapidly closing lag of only 3 to 6 months at up to 70%+ lower inference costs.

Frontier Model Families Capability Evolution - GPQA Benchmark (2023-2025)

Performance on Graduate-level Physics, Chemistry & Biology Question Answering



Llama



Qwen



Gemma



Mistral



DeepSeek



Phi



Molmo



Granite



Nemotron



Open Model Validation

Continuously benchmark, curate, and optimize open models to deliver peak performance without API lock-in.



Supply Chain Integrity

Enforce verifiable provenance using OCI (ModelPack), SPDX 3.0 AI BOMs, and Sigstore signatures for tamper-proof integrity.



Domain Customization

Leverage post-training, fine-tuning, and RL loops via Training Hubs to adapt foundation models to local regulations and custom data.

Sources:

[The Latent Role of Open Models in the AI Economy](#) (Frank Nagle and Daniel Yue, 18 November 2025)

[Sovereign Large Language Models: Advantages, Strategy and Regulations](#) (Bondarenko et al., 15 January 2025)

Trust Beyond Code: Auditable Supply Chains & Provenance

Applying proven software supply chain security principles directly to the model lifecycle



Model Openness Framework (MOF)

Designed by LF AI & Data, this tiered rubric evaluates model completeness, requiring open training datasets, pipelines, and evaluation metrics.



Licensing with OpenMDW 1.1

Eliminating intellectual property and legal ambiguity by applying OSI-approved code licenses alongside purpose-built weights and parameter terms.



Standardized & Portable Packaging

We package models as standardized OCI artifacts using the ModelPack standard, making them portable and manageable like any cloud-native application.



AI Bill of Materials (AI BOM)

Enforcing automated, machine-readable tracking of training lineages, exact datasets, known packages, and supply chain dependencies via SPDX 3.0 for verifiable provenance.



Cryptographic Integrity

Utilizing Sigstore and Notary v2 to generate immutable, tamper-evident transparency logs (Rekor) to secure model weights from code to production to guarantee model origin and integrity.





National AI Policy Case Study: Singapore

How Singapore aligns open-source testing, global tech partnerships, and pragmatic soft law to anchor trustworthy AI.



PILLAR 01

AI Sovereignty

Project SEA-LION: Open regional foundation LLM preserving Southeast Asian linguistic context.

Digital Trust Centre: Singapore AISI driving national evaluation testbeds.

Talent Density: Top-tier global ranking with 109.5 AI researchers per 100k citizens.



FEATURED INITIATIVE

AI Verify Foundation

Open Source

Global Consortium: Not-for-profit foundation uniting IMDA with Google, Microsoft, IBM, Salesforce, and Red Hat.

Technical Toolkits: Standalone open-source testing harness validating bias, fairness, and robustness.

Assurance Pilot: Developer-ready Starter Kits and red-teaming sandboxes for GenAI and LLM safety.



PILLAR 03

Governance & Safety

Model Frameworks: Progressive national guidance for Traditional, Generative, and Agentic AI.

Global Crosswalks: Direct interoperability mapping to NIST AI RMF and ISO/IEC 42001.

Regional Leadership: Foundational architecture for the multilateral ASEAN AI Governance Guide.

#1

IMF AI Preparedness Index

61%

National GenAI Adoption Rate

60+

Global Tech & Enterprise Members

Key Takeaway: Open ecosystem tooling converts voluntary governance principles into verifiable technical standards without prescriptive legislation.



National AI Policy Case Study: South Korea

Executing a "Dual Full-Stack" Strategy: Industrial Hardware Dominance vs. Structural Compute & Model Trade-offs

PILLAR 01

Dual Full-Stack Strategy

735B Initiative AI Framework Act K-AI Gov

\$735B Sovereign AI Initiative: Massive multi-year push allocating \$300B to AI data centers, \$185B to government R&D, and \$250B to industrial transformation (AX).

Regulatory & Safety Anchors: Enacted the APAC region's first omnibus AI legislation (AI Framework Act, Jan 2026) alongside the Presidential National AI Committee and Korea AI Safety Institute.

Public-Private Cloud PPPs: Modernizing public administration by deploying domestic LLMs across secure government cloud zones (NIS guidelines).

PILLAR 02

Infrastructure & Silicon

HBM Leader 50 Data Centers AI Factory

Memory Dominance vs. Logic Gap: Samsung and SK Hynix control global High-Bandwidth Memory (HBM3e/HBM4), but logic GPUs depend on NVIDIA, accelerating domestic NPU efforts.

Power & Infrastructure Bottlenecks: Building 50 new AI data centers (2GW+ capacity) is constrained by power grid connection delays (KEPCO) and cooling limits.

AI Factory Operating Models: Tensions between domestic CSP public-private cloud zones and mega-alliances like SK Telecom + NVIDIA (gigawatt-scale DSX AI Factory) and AWS + SK.

PILLAR 03

Sovereign Model Dilemma

Indigenous LLM From Scratch vs Open AX Focus

The "From Scratch" Friction: The MSIT Indigenous AI Foundation Model project required models pre-trained without open code reuse, causing major players (Naver Cloud, Kakao) to step aside.

Scratch Pre-Training vs. Open Weights: Debate over spending billions to pre-train proprietary LLMs from scratch versus post-training global open weights (Llama, Qwen, DeepSeek).

Localized Domain Customization: Transitioning toward domain-specific fine-tuning (industrial AX, legal, medical) to maximize value over raw AGI benchmark scaling.

#1

Global HBM Memory Share

\$735B

Total Public-Private Investment

Jan 2026

First APAC Omnibus AI Act

Key Takeaway: South Korea demonstrates that AI sovereignty is an industrial capability race. Achieving long-term autonomy requires 'managed interdependence'—leveraging domestic semiconductor strengths while forming pragmatic alliances and adopting open-weight foundation models.



South Korea's Sovereign AI Imperative: From Open Source Consumer to Global Co-Architect



The Sovereignty Pivot

OLD PARADIGM

"Open Source Consumer"

- Downstream user of foreign models & frameworks.
- Capital-intensive "from scratch" isolationism.
- Risk of "pulling up the ladder" behind Big Tech standards.

NEW PARADIGM

"Global Co-Architect"

- Active upstream contributor & specification leader.
- Leverages #1 HBM Memory share & \$735B AI initiative.
- Exercises "managed interdependence" via open ecosystems.



1. Compute & Hardware Layer (PyTorch Foundation)

PyTorch Core

vLLM

OpenReg

Triton DSL

- Upstream domestic NPU drivers (Rebellions, FuriosaAI) into PyTorch via OpenReg to break CUDA lock-in natively.
- Co-develop portable Triton/Helion attention kernels to run open weights across domestic and global silicon.



2. Cloud-Native Substrate Layer (CNCF)

Kubernetes

KServe

llm-d Routing

Envoy AI Gateway

- Shift operational control out of fragile Python app code into platform primitives.
- Drive standardized multi-tenant GPU isolation, zero-trust SPIFFE identity, and energy-aware data center orchestration.



3. Agentic Interoperability & Governance Layer (AAIF & LF)

AAIF / Linux Foundation

MCP

A2A Protocol

AI Safety

- Co-chair and contribute to open agentic standards (MCP for tool access, A2A for peer-agent routing) under neutral Linux Foundation governance.
- Align Korea AI Safety Institute evaluators with open, machine-readable testing frameworks (AI Verify) to export domestic trust models globally.

"True sovereignty is not achieved through isolation—it is forged by co-architecting the open standards, runtimes, and protocols that the world runs on."

Thank you

